INCH-POUND

MIL-C-24040B(SH)
19 January 1993
SUPERSEDING
MIL-C-24040A(EC)
19 March 1970
(See 6.13)

# MILITARY SPECIFICATION

# COMPUTER-INDICATOR (SCALER), RADIAC, CP-792( )/UD

*This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.*

## 1. SCOPE

**1.1 Scope.** This specification covers a militarized RADIAC Computer-Indicator (scaler) equipment for ship and shore use. The equipment is for use in computing and indicating the total number of random pulses received from external radioactivity detecting instruments, and for supplying energizing voltage to such instruments. Only one classification of equipment is covered by this specification.

## 2. APPLICABLE DOCUMENTS

### 2.1 Government documents.

**2.1.1 Specifications, standards, and handbooks.** The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-S-901 — Shock Tests, H.I. (High-Impact), Shipboard Machinery, Equipment, and Systems, Requirements for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6665

**DISTRIBUTION STATEMENT A:** Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:49Z
Check the source to verify that this is the current version before use.

MIL-C-3098 Crystal Units, Quartz, General Specification for

MIL-C-3098/4 Crystal Unit, Quartz, CR19A/U

MIL-F-15733 Filters and Capacitors, Radio Equipment Interference, General Specification for

MIL-E-17555 Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts); Packaging of

STANDARDS

MILITARY

MIL-STD-108 Definition of and Basic Requirement for Enclosure for Electrical and Electronic Equipment

MIL-STD-109 Quality Assurance Terms and Definitions

MIL-STD-167-1 Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)

MIL-STD-415 Test Provisions for Electronic Systems and Associated Equipment, Design Criteria for

MIL-STD-454 Standard General Requirements for Electronic Equipment

MIL-STD-461 Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference

MIL-STD-462 Electromagnetic Interference Characteristics, Measurement of

MIL-STD-471 Maintainability Verification/Demonstration/Evaluation

MIL-STD-781 Reliability Testing for Engineering Development, Qualification and Production

MIL-STD-785 Reliability Program for Systems and Equipment Development and Production

MIL-STD-1472 Human Engineering Design Criteria for Military Systems, Equipment and Facilities



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:49Z
Check the source to verify that this is the current version before use.

MIL-STD-2036 General Requirements for Electronic Equipment Specifications

HANDBOOKS

MILITARY

MIL-HDBK-217 Reliability Prediction of Electronic Equipment

MIL-HDBK-472 Maintainability Prediction

MIL-HDBK-718 Reliability Test Methods, Plans, and Environments for Engineering, Development, Qualification, and Production

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

**2.2 Non-Government publications.** The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

RS-310-C-77 Racks, Panels, and Associated Equipment. (DOD adopted)

(Application for copies should be addressed to Electronic Industries Association, 2001 Eye Street, NW, Washington, DC 20006.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

**2.3 Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

**3.1 First article.** When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.3.1.

**3.2 General requirements.** The equipment shall be in accordance with MIL-STD-2036, except as specified herein.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:49Z
Check the source to verify that this is the current version before use.

**3.2.1 Electronics.** Solid-state circuitry shall be employed unless otherwise approved by the command or agency concerned.

**3.2.2 Material.** Material shall be in accordance with MIL-STD-2036 except as otherwise specified or authorized herein.

**3.2.3 Parts.** All parts used in the construction of this equipment shall be selected and applied in accordance with MIL-STD-2036. No parts shall be used which require special selection from commercial production runs in order to meet equipment requirements. No mercury or mercury compound shall be used in this equipment. Parts shall be rated at plus 51.67 degrees Celsius (°C) (125 degrees Fahrenheit (°F), or higher if required by circuit application. Plastic encapsulated microcircuits shall not be used, unless approved by the command or agency responsible for non-standard parts approval (see 6.5.2).

**3.2.4 Shock.** The equipment shall withstand the shock tests for grade A, class I, type A, lightweight equipments in accordance with MIL-S-901 and shall be energized during the tests. After each blow, the equipment shall continue to operate without internal adjustments and without replacement of any part. There shall be no evidence of collision between any parts of the equipment.

**3.2.5 Vibration.** The equipment shall be subjected to the type I vibration test of MIL-STD-167-1 and shall be energized during the test. The instrument shall operate without internal adjustments and without replacements of any part. There shall be no evidence of collision between any parts of the equipment.

**3.2.6 Environmental conditions.** The equipment shall meet the temperature requirements of temperature range 4 and the humidity requirements of MIL-STD-2036.

**3.3 Quantitative reliability requirements.** The RADIAC Computer-Indicator (scaler) equipment shall have a specified mean-time-between-failure (MTBF) of 2000 hours. Other tasks entailed shall be in accordance with MIL-STD-785 and MIL-HDBK-217 (see 6.3).

**3.4 Component parts.** The equipment shall consist of the following component parts required to make up a complete equipment:

(a) One computer-indicator (scaler)

(b) One high-voltage cable 6 feet long, with UG-932-A connector at each end

(c) One power cable 6 feet long

(d) One maintenance extender board which fits any printed circuit board

**3.4.1 Computer indicator (scaler).** The equipment shall include input circuitry, amplifier, discriminator, scaling stages, stable oscillator, count indicators, as necessary to count random-pulses within the limits specified herein; plus a high-voltage power supply to energize external detectors.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:49Z
Check the source to verify that this is the current version before use.

**3.4.1.1 Human factors.** The contractor shall apply the human engineering principles of MIL-STD-1472 to the RADIAC Computer-Indicator (scaler) (see 6.3).

**3.4.1.2 Size and weight.** The equipment shall be rectangular with maximum dimensions of 17 inches width, 14 inches depth, and 7 inches height. The weight shall not exceed 50 pounds.

**3.4.1.3 Cabinet.** The equipment shall be housed in a cabinet of minimum size and weight consistent with other requirements. Shock and vibration resistance shall be attained without the use of shock mounts. The degree of enclosure shall be "dripproof" in accordance with MIL-STD-108. The form factor and arrangement of controls, connectors, and indicators shall be subject to the approval of the command or agency concerned (see 6.3).

**3.4.1.3.1 Front cover.** The front cover shall be removable for maintenance without removing or disturbing adjacent parts or equipment.

**3.4.1.4 Housing surface.** The housing surface shall have a smooth surface with a minimum of crevices and sharp edges. Rubber or felt feet shall not be permitted on the case. All indication marking scales shall be engraved on the plainly marked, smooth surface of the case.

**3.4.1.5 Chassis.** The chassis shall be removable from the cabinet and mounted in a standard 19 inch electrical equipment rack in accordance with EIA RS-310-C-77. The chassis shall be on slides for removal from the cabinet housing. The safety requirements of MIL-STD-2036 shall apply.

**3.5 Controls.**

**3.5.1 Control panels.** The controls, indicators, and connectors shall be located as follows:

(a) *Front panel:*

(1) Switch, master
(2) Pilot light, master
(3) Manual switch or keypad, count and time
(4) Switch or keypad, preset time selector
(5) Indicator, counts accumulated
(6) Indicator, time of count
(7) Switch, high-voltage
(8) Pilot light, high-voltage
(9) Control knob or keypad, high-voltage
(10) Indicating meter or light emitting diode (LED) display, high-voltage
(11) Ten-turn potentiometer or keypad discriminator calibration adjust.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:49Z
Check the source to verify that this is the current version before use.

(b) Rear panel:

(1) Input sensitivity adjust
(2) Adjustments for internal voltage
(3) Connection, primary power
(4) Connector counter input, with high-voltage. Connector shall be rated at 2500 volts (V).
(5) Fuses, spare fuses, and blown fuse indicators
(6) Test points, of the type designated "accessible" in accordance with MIL-STD-415
(7) Power of preamplifier. Connector shall be rated at 2500 volts.

**3.5.1.1 Internally located controls.** Adjustments for input sensitivity and internal voltages, spare fuses, and test points shall be located internally and shall be subject to the approval of the command or agency concerned (see 6.3).

**3.5.1.2 Accessibility of internally located controls.** All internally located controls and adjustments shall be positioned to be adjustable when the equipment is energized for use or calibration and the results of these adjustments shall be readable directly during adjustment.

**3.5.2 Discriminator adjust control.** The discriminator adjust control shall be designed to permit positive and negative pulses of .25 and 20 volts peak amplitude to be selected, thereby allowing all pulses between (positive) +10 volts and (negative) −65 volts peak amplitude to be counted on a one-to-one basis. High-level input pulses shall not produce double pulses or no pulses. This control shall be a ten-turn precision potentiometer with dual dials to indicate the relative position of its contact arm: the outer dial shall show hundredths of units from 1 through 10, and the inner dial shall show units up through 99. Together the dials shall provide 1000 adjustment index points. Calibration adjustment shall be accomplished through this method or by entering discriminator calibration adjustment setpoints through a front panel keypad.

**3.5.3 Locking devices.** Controls on the front panel shall not have locking devices, except for the discriminator adjust. The discriminator shall be locked-pot type configuration to ensure positive and accurate setting. Controls on the rear panels shall have locking devices. If the keypad calibration adjustment is used, no locking device is required on the front panel; however, a keypad function shall exist that maintains and ensures a positive and accurate setting.

**3.5.4 Count indicator.** The count indicator shall consist of six decades, capable of accumulating and presenting a count of 999,999 in the decimal system. The indicating shall be accomplished with LED display. The LED display shall be capable of being dimmed electronically. The display equipment shall be presented to the command or agency responsible for non-standard parts for approval.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:49Z
Check the source to verify that this is the current version before use.

**3.5.5 Timer.** A timer system shall be provided to measure and indicate the time during which a count is made. The basic time reference shall be the internal frequency source. Time shall be indicated in decimal units of minutes up to 999.99 minutes with error in the overall indication of not more than (plus) +1 unit in the least significant decimal place. No other timers shall be furnished.

**3.5.5.1 Present times.** A control shall be provided to preset the intended counting time, so that the counter and timer shall stop automatically and simultaneously at the end of the preset time. Preset time shall be adjustable to the maximum timing indication (see 3.5.5). The count and time indicators shall retain their indications after a counting run has been stopped until the indicators are reset, or the equipment turned off. It shall not be possible to start a new count without resetting time and count indicators to zero. The timer shall have quartz crystal controlled internal stable oscillator as required by MIL-C-3098/4 with accuracy as required (see 3.5.2).

**3.5.5.2 Internal frequency source.** An internal stable oscillator shall control the test signal and timer. The oscillator shall be accurate to at least 1 part in $10^4$ seconds.

**3.5.6 Test signal.** The equipment shall include a circuit to furnish a pulsed test signal at 100 Hertz (Hz) to actuate the scaling circuit. The unshaped signal shall be obtained from the stable frequency source used to drive the timer (see 3.4.5.2). This test signal shall be applied to the input of the instrument (including the input amplifier and discriminator) with an amplitude of 250 millivolts (mV) to produce a desired display of 6000 ± 1 counts. The output reading is reference to the timer oscillator.

**3.5.7 Power supply control.** Control shall be by means of a ten-turn calibrated potentiometer or through keypad function and high-voltage switch.

**3.5.7.1 Discriminator modes.** Equipment design shall allow the discriminator mode to be adjusted by a three-position switch or keypad located on the front panel. The first mode of the discriminator switch would automatically set the discriminator to −1/4 volt (GM tube) setting. The second mode would automatically set the discriminator to the desired (calibration) setting. The third mode would provide for the present ability of (continuous) setting by a ten-turn potentiometer over the specified voltage range. The switch settings shall be marked as indicated within the above parentheses. If the keypad function is used, the discriminator mode shall be through keypad selections. The discriminator mode shall have GM tube, calibration, and continuous selections as described above.

**3.5.7.2 High-voltage ranges.** Equipment design shall allow the high voltage to be adjusted to two ranges. The first range shall cover 0 to 1600 volts and the second range shall cover 0 to 2500 volts. Range adjustment shall be by an internal switch setting. A front panel LED display shall



Source: https://assist.dla.mil -- Downloaded: 202610-19T18:49Z
Check the source to verify that this is the currentversion before use.

indicate to which high-voltage range the RADIAC Computer-Indicator is internally set. The red LED display shall have two positions. The first position shall correspond to the 0 to 1600 volt range and the second position shall correspond to the 0 to 2500 volt range. The switch shall be factory set to the 0 to 1600 volt range.

**3.6 Power.**

**3.6.1 Primary power.** The equipment shall be designed to operate from a supply line of 115 volts, single-phase, alternating current, and 60 Hertz.

**3.6.2 Power line isolation.** Power line conducted noise (external noise) shall be suppressed to prevent interference with the count. Both legs of the power line shall be filtered with a filter in accordance with MIL-F-15733. The filter shall have an insertion loss of 50 decibels (db) or greater for frequencies between 0.15 megahertz (MHz) and 40 MHz. The power transformer shall contain a grounded electrostatic shield. The scaler shall be fused on the line side of the filter such that shipboard fault circuit alarms shall not be activated.

**3.6.3 High-voltage power supply.** A high voltage power supply shall be included to provide voltage for the operation of external detectors.

**3.6.3.1 Voltage range.** The voltage range shall be continuously adjustable from +500 volts to +2500 volts.

**3.6.3.2 Stability.** Variation shall be not greater than plus or minus 1/2 percent (±0.5%) at all voltage settings, after warm-up, over a period of one hour.

**3.6.3.3 Regulation.** Regulation shall be as follows:

(a) *Line.* Not greater than one percent for voltage change of 10 percent or frequency of 5 percent, in primary input power.

(b) *Load.* Not greater than one percent for variation in load between zero and 1/4 milliampere (0-0.25 ma) over the full range.

**3.6.3.4 Indication.** The high-voltage indication shall include the following:

(a) A pilot light which shall be "ON" when the switch is "ON".

(b) An indicating meter with scale length of 2.5 inches or digital LED display. The meter shall read from 0 to 2500 volts in 50 volt scale divisions. The digital display shall be used to read from 0 to 2500 volts.

(c) The 2500 volt panel meter shall indicate the high voltage with a maximum error of ±2 percent. The digital display shall indicate the high voltage with a maximum error of ±2 percent.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:49Z
Check the source to verify that this is the current version before use.

**3.6.3.5 Short-circuit protection.** Short-circuit protection shall be provided for the high-voltage supply.

**3.6.4 Preamplifier power.** All regulated voltages developed internally are to appear at one multi-pin connector at the rear panel power for the preamplifier.

**3.7 Operating characteristics.**

**3.7.1 Scaler.** The scaler shall consist of scaling stages and such associated circuitry as required to furnish decimal counts to the count indicator. The CP-792 input circuitry shall pulse count the output signal waveforms of the Beta counting system, shown on figure 1; and the AN/PDR-56 RADIAC set, shown on figure 2, on a one-for-one basis.

**3.7.1.1 Signal resolution.** The equipment shall resolve individual pulses separated by not more than 1 microsecond.

**3.7.2 Accuracy.** The counter accuracy shall be within plus or minus one count of the total number of counts accepted by the scaler. A discriminator voltage setting of −0.25 volts shall be used to check equipment accuracy.

**3.7.3 Input impedance.** The input impedance shall be one megohm over the frequency range 50 Hz to 5 MHz. Signal input impedance shall not be less than one megohm at any time of operation of the circuit.

**3.7.4 Noise rejection.** The equipment shall reject all pulses of less than 0.25 volts peak amplitude, and be consistent with 3.5.2.

**3.7.5 Shock.** The equipment shall operate without adjustment or replacement of parts during the shock requirements as specified in 3.2.4.

**3.7.6 Vibration.** The equipment shall operate without adjustments or replacement of parts as specified in the vibration requirements of 3.2.5.

**3.7.7 Environment.** The equipment shall operate within the temperature and humidity requirements as specified in 3.2.6.

**3.7.8 Controls.** Controls shall include those listed and as specified in 3.5.1 through 3.5.6.

**3.7.9 Power.** The equipment shall operate with the power requirements as specified in 3.6 through 3.6.3.4.

**3.7.10 Electromagnetic compatibility.** The scaler set shall meet the requirements of MIL-STD-461, part 5 (class A4), tests RE01, RE02, RS01, RS02, and RS03. The limits for test RS03 shall be as specified in table I.



Source: https://assist.dla.mil -- Downloaded: 202640-19T18:49Z
Check the source to verify that this is the currentversion before use.

TABLE I. *Frequency and field intensity for electromagnetic campatibility.*

| Frequency (MegaHertz (MHz)) | Field Intensity (Volt per meter (V/m)) |
|---|---|
| 0.014 - 0.30 | 1.0 Average |
| 0.30 - 1.5* | 1.0 Average |
| 1.5 - 30* | 10 Average |
| 30 - 12,400 | 1.0 Average |

*Test at only three discrete frequencies within these ranges.

**3.7.11 Reliability.** The equipment shall have a mean-time-between failure (MTBF) of 2000 hours and meet the requirements as specified in 4.5 through 4.5.4.

**3.7.12 Safety.** The equipment shall operate within the safety design criteria of MIL-E-16400 (*safety criteria*) and MIL-STD-454, as applicable.

**3.7.12.1 Safety interlock.** The equipment shall have a safety interlock which conforms to MIL-E-16400 (interlock indicator) and MIL-STD-454, requirement 1, 5.7.c (see 6.3).

## 4. QUALITY ASSURANCE PROVISIONS

**4.1 Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

**4.1.1 Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:49Z
Check the source to verify that this is the current version before use.

**4.1.2 Quality program.** (See 6.3).

**4.1.2.1 Quality assurance terms and definitions.** Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

**4.1.3 Changes.** Any changes to the approved plan shall be submitted to the Quality Assurance Representative for approval prior to use.

**4.1.4 Government verification.** All quality assurance operations performed shall be subject to Government verification at any time. Verification shall consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written system description are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with all the requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance.

**4.1.5 Test procedures.** (See 6.3).

**4.2 Classification of inspections.** The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

**4.3 First article inspection.** Unless otherwise specified, three units shall be required for first article inspection. First article inspection shall consist of the examination and testing necessary to determine compliance with this specification and MIL-STD-2036, using the examination and tests of MIL-STD-2036 as specified in 4.3.1.

**4.3.1 First article examination and tests.** The following examination and tests of MIL-STD-2036 shall be performed: (Unless otherwise noted, refer to examination and tests table in MIL-STD-2036.)

(a) Examination

(b) Performance

(c) Temperature - (low) (Perform using sheltered equipment - Procedure II, basic cold.) (Temperature range 4)

(d) Temperature - (high) (Perform with 50 °C high temperature range) (Temperature range 4)

(e) Humidity



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:49Z
Check the source to verify that this is the current version before use.

(f) Salt fog

(g) Shock (may be performed on one unit only)

(h) Vibration

(i) Enclosure (to follow shock and vibration tests)

(j) Steady-state voltage and frequency

(k) Transient voltage

(l) Transient frequency

(m) Spike voltage

(n) Power interruption

(o) Electromagnetic environmental effects (includes tests RE01, RE02, RS01, RS02, and RS03 of MIL-STD-462) (see table III for RS03)

(p) Inclination (including submarine requirement).

**4.3.1.1 Non-standard parts.** First article approval shall not be made until all non-standard parts have been approved. Germanium semi-conductors are not acceptable as non-standard parts for this equipment. Non-standard parts shall be approved in accordance with MIL-STD-2036 (parts) (see 6.5.2).

**4.4 Quality conformance inspection.** The quality conformance inspection shall consist of the following examination and tests. (Refer to examination and tests table in MIL-STD-2036 unless otherwise noted.)

(a) Examination

(b) Performance

**4.4.1 Production quality conformance sampling.** Production quality conformance sampling shall be in accordance with MIL-STD-2036 and table II. Production quality conformance sampling shall be performed on equipments that have passed quality conformance inspection, and shall consist of the following examination and tests. (Refer to examination and tests table in MIL-STD-2036 unless otherwise noted.)

(a) Low temperature (Perform using sheltered equipment - Procedure II, basic cold) (Temperature range 4)

(b) High temperature (Perform with plus 50 °C high temperature range) (Temperature range 4)

(c) Enclosure



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:49Z
Check the source to verify that this is the current version before use.

(d) Steady-state voltage and frequency

(e) Power interruption

(f) Electromagnetic environmental effects (CE06 not required).

TABLE II. Sampling for production control inspection.

| Major defects | | | |
|---|---|---|---|
| Lot size | Sample size | Accept | Reject 1/ 2/ 3/ |
| 2 - 50 | 2 | 0 | 1 |
| 51 - 500 | 8 | 1 | 2 |
| 501 - 3200 | 13 | 2 | 3 |
| 3201 - 35000 | 20 | 3 | 4 |
| Minor defects | | | |
| 2 - 150 | 5 | 1 | 2 |
| 151 - 500 | 8 | 2 | 3 |
| 501 - 3200 | 13 | 3 | 5 |
| 3201 - 35000 | 20 | 5 | 6 |

1/ All defective items shall be replaced with acceptable items prior to lot acceptance.

2/ Inspect sample size until reject criteria is reached. If reject criteria is reached, reject the entire lot.

3/ Rejected lots may be screened and resubmitted for inspection and retest.

**4.4.2 Design requalification inspection.** Design requalification inspections shall be conducted when the design of, or material used in, the equipment and such change may affect the equipment's ability to comply with one or more of the environmental test requirements. Design requalification inspections shall consist of the following tests in accordance with MIL-STD-2036. (Refer to examination and tests table in MIL-STD-2036 unless otherwise noted.)

(a) Electromagnetic environmental effects (includes tests RE01, RE02, RS01, RS02, and RS03 of MIL-STD-462) (see table III for RS03)

(b) Spike voltage

(c) Transient frequency



Source: https://assist.dla.mil -- Downloaded: 202640-19T18:49Z
Check the source to verify that this is the currentversion before use.

(d) Transient voltage

(e) Inclination (including submarine requirement)

(f) Humidity

(g) Salt fog

(h) Shock (may be performed on one unit only)

(i) Vibration.

**4.5 Reliability.**

**4.5.1 Reliability prediction.** A design reliability prediction shall be performed in accordance with MIL-HDBK-217 (see 6.3). The results of the prediction shall be compared to the specified MTBF and where the predicted values do not achieve the requirements, such changes shall be accomplished as are necessary to increase the predicted values to achieve the requirements.

**4.5.2 Reliability testing.** The reliability of the RADIAC Computer-Indicator (scaler) shall be demonstrated on the initial production units (see 6.3). Failure shall be defined as any malfunction or parameter deviation that prevents the equipment from performing within the requirements set forth in this specification. Debugging and burn-in prior to the reliability test shall be limited to the debugging and burn-in to be performed on all production equipment prior to delivery to the Government. No preventative maintenance shall be performed during the reliability test. Reliability testing shall include vibration tests.

**4.5.3 Performance parameters for reliability testing.** The parameters to be monitored shall include all of the requirements specified in section 3 herein.

**4.5.4 Reliability tests.** Reliability testing shall be conducted on the initial production units in accordance with MIL-STD-781 and MIL-HDBK-718. Testing will be performed as specified in test plan IV with the following test conditions:

| | |
|---|---|
| Temperature | 50 +5, −0 °C (122 to 131 °F) |
| Temperature Cycling | None |
| Vibration | 2.2G ± 10 percent peak acceleration at any nonresonant frequency between 20 and 60 cps measured at the mounting points on the equipment. (Equipment designed for use on Navy ships shall be vibrated at 1G ± 10 percent. If the equipment is designed to meet a vibration requirement less severe than the requirement contained herein, the vibration may be reduced to the level specified as a design requirement.) The duration of vibration shall be at least 10 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:49Z
Check the source to verify that this is the current version before use.

| | |
|---|---|
| | minutes during each hour of equipment operating time. When the equipment being tested contains circuit boards or cards, vibration shall be normal to the plane of the majority of the cards. This vibration requirement may be met by mounting the equipment solidly (that is, without the equipment shock vibration mounts) to a strong flat plate which is supported by vibration isolators. The desired amplitude and frequency may be obtained by a simple vibration machine with an asymmetric weight on a shaft attached to the plane and driven by a motor. It is recommended that the vibration equipment be checked for proper operation each 24 hours of operation, and that the vibration transducer be "ON" and monitored continuously during vibration. |
| Equipment On-Off Cycling | Turn on and let temperature stabilize, hold for 3 hours, then turn off and let temperature stabilize. This cycle shall continue throughout the test. |
| Input Voltage | Nominal specified voltage +5 −2 percent. |
| Input Voltage Cycling | When so directed by the contracting activity, voltage cycling shall be accomplished as follows: The input voltage shall be maintained as 110 percent nominal for one-third of the equipment "ON" cycle, and at 90 percent for the final one-third of the equipment "ON" cycle. This cycling procedure is to be repeated continuously throughout the reliability test. |

Testing shall be performed on four (4) production units. Reliability testing shall be successfully completed prior to equipment shipment.

**4.5.5** ***Reliability sampling tests.*** Reliability sampling tests shall be conducted throughout production to ensure continued production of acceptable equipment. Testing shall be performed as specified in test plan IV with the following conditions:

| | |
|---|---|
| Temperature | 50 +5 −0 °C (122 to 131 °F) |
| Temperature Cycling | None |
| Vibration | 2.2G ± 10 percent peak acceleration at any nonresonant frequency between 20 and 60 cps measured at the mounting points on the equipment. (Equipment designed for use on Navy ships shall be vibrated at 1G ± 10 percent. If the equipment is designed to meet a vibration requirement less severe than the requirement contained herein, the vibration |



Source: https://assist.dla.mil -- Downloaded: 202640-19T18:49Z
Check the source to verify that this is the currentversion before use.

may be reduced to the level specified as a design requirement.) The duration of vibration shall be at least 10 minutes during each hour of equipment operating time. When the equipment being tested contains circuit boards or cards, vibration shall be normal to the plane of the majority of the cards. This vibration requirement may be met by mounting the equipment solidly (that is, without the equipment shock vibration mounts) to a strong flat plate which is supported by vibration isolators. The desired amplitude and frequency may be obtained by a simple vibration machine with an asymmetric weight on a shaft attached to the plane and driven by a motor. It is recommended that the vibration equipment be checked for proper operation each 24 hours of operation, and that the vibration transducer be "ON" and monitored continuously during vibration.

| | |
|---|---|
| Equipment On-Off Cycling | Turn on and let temperature stabilize, hold for 3 hours, then turn off and let temperature stabilize. This cycle shall continue throughout the test. |
| Input Voltage | Nominal specified voltage +5 −2 percent. |
| Input Voltage Cycling | When so directed by the contracting activity, voltage cycling shall be accomplished as follows: The input voltage shall be maintained as 110 percent nominal for one-third of the equipment "ON" cycle, and at 90 percent for the final one-third of the equipment "ON" cycle. This cycling procedure is to be repeated continuously throughout the reliability test. |

MIL-STD-781 and MIL-HDBK-718 shall be used to determine the number of units to be selected for reliability sampling tests. Reliability testing for each lot shall be successfully completed prior to equipment shipment.

**4.6 Maintainability.**

**4.6.1 Quantitative maintainability requirements.** The RADIAC Computer-Indicator (scaler) shall have an equipment mean-time-to-repair (MTTR) of not greater than 30 minutes (see 6.3 and 6.11).

**4.6.2 Maintainability demonstration.** The maintainability requirements for the equipment shall be successfully demonstrated in accordance with method III of MIL-STD-471. The maintainability test shall be performed on initial production units (see 6.3).



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:49Z
Check the source to verify that this is the current version before use.

**4.7 Inspection of packaging.** Sample packages and packs, and the inspection of the preservation, packing and marking for shipment, stowage, and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

## 5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2 (see 6.8).)

**5.1 Preservation, packing, and marking.** Unless otherwise specified, packaging shall be in accordance with MIL-E-17555 (see 6.2).

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

**6.1 Intended use.** The RADIAC, CP-792( )/UD, Computer-Indicator (scaler), is for use in computing and indicating the total number of random pulses received from external radioactivity detecting instruments and for supplying energizing voltage to such instruments.

**6.2 Acquisition requirements.** Acquisition documents must specify the following:

(a) Title, number, and date of this specification

(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)

(c) First article inspection, if required (see 3.1 and 4.3)

(d) When technical manuals are required (see 6.4)

(e) When panel design must be approved prior to production (see 3.4.1.1)

(f) Number of first article samples to be submitted if other than specified in 4.3

(g) Levels of preservation, packing, and marking required (see 5.1).

**6.3 Consideration of data requirements.** The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.



Source: https://assist.dla.mil -- Downloaded: 202610-19T18:49Z
Check the source to verify that this is the currentversion before use.

| Reference Paragraph | DID Number | DID Title | Suggested Tailoring |
|---|---|---|---|
| 3.4.1, 3.4.1.3, 3.5.1.1, and 3.7.12.1 | DI-HFAC-80242 | Human factors design analysis report | – |
| 4.1.1 | DI-NDTI-80809 | Test, inspection reports | - |
| 4.1.2 | DI-R-5297 | Quality program plan (QPP) | – |
| 4.3 | DI-NDTI-80604 | Test procedures | Subtitle CDRL "First article inspection procedures |
| 4.3 | DI-NDTI-80604 | Test report | Subtitle CDRL "First article inspection report |
| 4.5.1 | DI-R-7082 | Reliability predictions report | – |
| 4.5.2 | DI-RELI-80250 | Reliability test plan | – |
| 4.5.2 | DI-RELI-80252 | Reliability test reports | – |
| 4.6.1 | DI-MNTY-80822 | Maintainability program plan | – |
| 4.6.1 | DI-MNTY-80827 | Maintainability predictions report | – |
| 4.6.1 | DI-MNTY-80828 | Maintainability analysis report | – |
| 4.6.2 | DI-MNTY-80831 | Maintainability demonstration test plan | – |
| 4.6.2 | DI-MNTY-80832 | Maintainability demonstration report | – |

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

**6.4 Technical manuals.** The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List, which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.



Source: https://assist.dla.mil -- Downloaded: 202640-19T18:49Z
Check the source to verify that this is the currentversion before use.

**6.5 First article.** When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

**6.5.1 Tests, test equipment, and test procedures.** Test procedures should include test equipment and tolerance limits to be used for first article inspection and quality conformance inspection. Written test procedures should be as specified in 6.3 herein.

**6.5.2 Non-standard parts.** The first article test report should include a list of non-standard parts approved or pending approval for the equipment.

**6.6 Provisioning.** Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

**6.7** When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

**6.8 Sub-contracted material and parts.** The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

**6.9 Production quality conformance sampling and a suggested acceptable quality level (AQL).** An AQL for production quality conformance sampling is 6.5 percent.

**6.10 Reliability block diagrams.** A functional block diagram and reliability block diagram should be submitted with the reliability prediction report. Mechanical parts should be included in the prediction as shown in accordance with MIL-HDBK-217. The source and applicability of all failure data used for the prediction should be documented and justified prior to use.

**6.11 Maintainability prediction.** MIL-HDBK-472 should serve as a basis for maintainability prediction data.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:49Z
Check the source to verify that this is the current version before use.

**6.12 Subject term (key word) listing.**

Control panel
External detectors
High-voltage
Radioactive pulses
Radioactivity detection

**6.13 Changes from previous issue.** Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6665-N587)



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:49Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER<br>MIL-C-24040B(SH) | 2. DOCUMENT DATE (YYMMDD)<br>93-01-19 |
|---|---|---|

3. DOCUMENT TITLE
COMPUTER-INDICATOR (SCALER), RADIAC, CP-792()/UD

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME *(Last, First, Middle Initial)* | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS *(Include Zip Code)* | d. TELEPHONE *(Include Area Code)*<br>(1) Commercial<br>(2) AUTOVON *(If applicable)* | 7. DATE SUBMITTED *(YYMMDD)* |

8. PREPARING ACTIVITY

| a. NAME Technical Point of Contact (TPOC):<br>Mr. Rhodes (04RL)<br>PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS: | b. TELEPHONE *(Include Area Code)*<br>(1) Commercial: (703) 602-2753<br>(2) AUTOVON: 8-332-2753 |
|---|---|
| c. ADDRESS *(Include Zip Code)*<br>Commander<br>Naval Sea Systems Command (SEA 05Q42)<br>2531 National Center Bldg 3<br>Washington, DC 20362-5160 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 |

Source: https://assist.dla.mil -- Downloaded: 2026-01-19T18:49Z
Check the source to verify that this is the current version before use.